$$\mu = \frac{1}{|S|} \sum_{s \epsilon S} s$$

Once defined the mean, we can then define the variance (usually denoted as σ^2) as the mean of the squares of the differences between `S`'s mean and each of its elements:

$$\sigma^2 = \frac{1}{|S|} \sum_{s \epsilon S} (s - \mu)^2$$

So, given a set of `n` points `P0..Pn-1`, each `Pj` with coordinates `(P(j,0), P(j,1), .., P(j,k-1))`, the formulas for mean and variance along the direction of the `i`-th axis are:

$$\mu_i = \frac{1}{n} \sum_{j=0}^{n-1} P_{j,i}$$

$$\sigma_i^2 = \frac{1}{n} \sum_{j=0}^{n-1} (P_{j,i} - \mu_i)^2$$

These formulas are easily translatable into code and, even more, in most programming languages you will find an implementation of the method computing variance in core libraries; therefore, we won't show the pseudocode here: let's see, instead, how both functions for variance and mean are used in method `updateBoundingEnvelope` (listing 10.10), that computes a node centroid and radius.